

MAINS MONITOR

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Conditions like heavy load and line faults can cause the mains supply to sag or surge without warning. This can damage the appliances and equipment powered by the mains supply, unless these are automatically switched off by a protection device like the mains monitor presented here.

This mains monitor continuously monitors the mains voltage and immediately disconnects power to the load if the mains voltage goes beyond the predefined limit. As the mains power returns to its normal voltage, it reconnects the load after a preset time delay. Also, it tells whether a mains disruption occurred while you were out. The acceptable voltage range and the time taken before the power is restored to the load can be set by the user.

Circuit description

Fig. 1 shows the mains monitor circuit. The circuit is built around LM3914 dot/bar LED driver (IC1), CD4060 timer/counter (IC2), CD4093 quad Schmitt NAND gate (IC3) and a few discrete components. At the heart of the circuit is LM3917 dot/bar display driver, which drives ten LEDs in response to the input DC voltage from the secondary coil of transformer X1. This input voltage changes in proportion to the mains voltage (say, 240V). The dot-mode LED display gives a direct readout of the incoming mains voltage.

The logic output of IC1 in response to the incoming voltage is used to control timer CD4060 (IC2) and relay RL1. IC CD4093 functions as an event memory latch with NAND gates N1 and N2. A 1Hz oscillator is built around NAND gate N3.

To derive the power supply for the circuit, the 230V AC mains is stepped

down by transformer X1 to deliver a secondary output of 15V-0-15V, 1A. The transformer output is rectified by two separate half-wave rectifier diodes D11 and D12. The rectified output from diode D11 is filtered by capacitor C9 and regulated by IC 7812 (IC4). Capacitor C8 bypasses the ripples present in the regulated supply. The filter stage made of resistor R16 and capacitor C7 provides a clean DC supply to the circuit.

The rectified output of diode D12 is used to monitor the mains voltage level. It is filtered by capacitor C6 and applied to an adjustable voltage divider formed by resistor R15, preset

PARTS LIST

Semiconductors:	
IC1	- LM3914 dot/bar LED driver
IC2	- CD4060 counter/timer
IC3	- CD4093 quad Schmitt NAND gate
IC4	- 7812, 12V regulator
T1	- BC337 npn transistor
D1-D9	- 1N4148 switching diode
D10-D12	- 1N4007 rectifier diode
LED1-LED12	- 3mm LED
Resistors (all ¼-watt, 45% carbon)	
R1, R2	- 1-kilo-ohm
R3	- 3.9-kilo-ohm
R4	- 180-kilo-ohm
R5	- 27-kilo-ohm
R6, R11, R12	- 1.8-kilo-ohm
R7	- 270-kilo-ohm
R8	- 1.2-mega-ohm
R9	- 10-kilo-ohm
R10	- 330-kilo-ohm
R13, R15	- 18-kilo-ohm
R14	- 8.2-kilo-ohm
R16	- 15-ohm
Capacitors:	
C1	- 10nF ceramic disk
C2, C8	- 0.1µF ceramic disk
C3	- 0.22µF ceramic disk
C4	- 3.3µF, 35V electrolytic
C5	- 10µF, 16V electrolytic
C6	- 47µF, 50V electrolytic
C7	- 3.3µF, 35V electrolytic
C9	- 1000µF, 50V electrolytic
Miscellaneous:	
X1	- 230V AC primary to 15V-0-15V, 1A transformer
RL1	- 12V, 1 C/O relay
PZ1	- Piezobuzzer
S1	- 8-way DIP switch
S2	- 4-way DIP switch
S3	- Push-to-on switch



VR1 and resistor R14. The adjustable voltage divider output is filtered by capacitor C5 and fed to input pin 5 of IC1. The voltage divider formed by resistors R1 through R4 makes IC1 act as a voltmeter. LED1 through LED10 glow as the input voltage at pin 5 of IC1 increments. Using resistors R1 through R4, IC1 is set to display 215V AC and 260V AC on LED1 and LED10, respectively. Pin 9 of IC1 is left open to make it run in 'dot' mode. Resistor R6 limits the current through LEDs.

Pins 11 through 18 of IC1 are coupled to diodes D2 through D9. Diodes D2 through D9, DIP switch S1 and pull-up resistor R5 form an OR gate at the junction of resistor R5 and DIP switch S1. If the voltage-range selector DIP switches connected to D5, D6 and D7 are closed, the OR gate output will be low and LED5, LED6 and LED7 glow to indicate that the mains voltage ranges between 235V and 245V—the acceptable voltage range. Conversely, the OR gate output goes high when the mains voltage is outside this range, as one of the other LM3914 outputs will be low. This OR gate output controls reset pin 12 of timer IC2 and LED12 through NAND gate N4.

Timer IC2 is enabled when mains voltage is within the predefined range but resets when the mains voltage goes out of the acceptable limit. Resistors R7 and R8, and capacitor C3 run the internal oscillator of IC2 so that outputs Q7, Q9, Q11 and Q13 provide a time delay of 15 seconds, 1 minute, 4 minutes and 16 minutes, respectively.

These four outputs are applied to hold-off delay switch S2 (DIP switch), which selects any one input and provides it to diode D1. Assuming DIP switch S2 selects Q9 output of IC2, Q9 output goes high after a de-